PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 06 – Feature Engineering: Transformation, Construction, and Selection

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Three periods follow the syllabus allocation

Allocation	Learning content
Transformation 45 min	Missing values, scales, nonlinear transforms and categories.
Construction 22.5 min	Distance, ratios, temporal summaries and feature interactions.
Selection 67.5 min	Filters, wrappers, embedded methods and a valid fitted pipeline.

Period arrangement

Period 2 contains construction and filters. Period 3 contains subset search, embedded selection and integrated exercises.

Learning goal

Given a raw row, calculate the resulting vector, explain each column and say which training data learned each rule.

The representation changes what a learner can use

Raw data

Measurements with units, categorical strings, timestamps, text and missing entries.

Model input

A consistent numeric matrix $X \in \mathbb{R}^{n \times p}$ and a target vector y. One row is one prediction case.

Same numbers, different meaning

A ward code of 3 is an identifier. A severity grade of 3 has an order. A temperature of 3 degrees has a numeric difference and unit.

Three operations

Transformation changes how a variable is represented. Construction combines available information into a new variable. Selection keeps a subset of existing columns.

Learning question

A convenient representation can help a model. It does not add information that was never observed.

Rows identify prediction cases and columns identify measurements

A raw table

Three admission records contain age, lab and ward. The numeric part has shape 3×2, while the categorical column requires a separate encoding.

After engineering

Two numeric columns, one missing indicator and two learned ward indicators give $X \in \mathbb{R}^{3\times 5}$.

One record

The third row becomes one five-coordinate vector, not five independent patients. Adding columns changes representation dimension, not sample count.

Prediction unit

Repeated visits from one person may need grouped or time-based splitting. A random row split can let related records appear on both sides.

One small table will anchor the calculations

Row	Age(years)	Lab(mg/L)	Ward	y
A	20	1	A	0
B	40	3	B	1
C	60	missing	A	1

Prediction task

Use measurements available at admission to predict a later outcome. Training has 3 rows. A new row is age 50, lab missing, ward C.

Questions

What replaces the missing lab? Which age mean and scale apply to the new row? How do we encode an unseen ward?

Availability

Discharge status is measured after the prediction time and cannot be an admission feature. A column can contain valid values yet be invalid for the task.

Fit learns a rule, transform applies that rule

Fit	Learn medians, centres, scales, categories or selected columns from training rows.
Transform	Apply these saved parameters to new rows with the same column order.

Age example

Training ages 20, 40, 60 have mean 40. New age 50 is centred as $50 - 40 = 10$, even if the new batch has mean 50.

Cross-validation

Each fold fits preprocessing on its training rows and transforms its validation rows. A pipeline preserves this order.

Leakage

Fitting a scaler or supervised selector on all rows before validation lets validation information influence training.

Refitting the test mean can reverse a centred sign

Training coordinate

Ages 20, 40, 60 give mean 40. A new age 50 should be centred to $50 - 40 = 10$.

Incorrect new-batch fit

The new batch contains ages 50, 70 with mean 60. Recentring that batch would encode age 50 as $50 - 60 = -10$.

Consequences

The same person's transformed sign changes because of the other batch member. A classifier trained in the mean 40 coordinate system receives a different representation.

Correct use

Save the training mean and scale, then apply them to every future batch. A genuine model-update policy is a separate fitted procedure requiring evaluation.

Missing values first require a useful explanation

Observed pattern	Consequence for treatment
Occasional sensor failure	Dropping a few rows may mostly lose data.
One ward seldom orders a test	Complete-case deletion changes the ward mixture.
Very ill patients cannot finish a test	Missingness may depend on the outcome-related state.

Inspect

Count missingness by column and relevant groups. Distinguish unavailable, unmeasured and invalid values.

Mechanisms

MCAR, MAR and MNAR describe assumptions about why data are missing. The observed table alone usually cannot prove the mechanism.

Task relevance

A measurement never available at prediction time should not become usable merely through imputation.

Deleting missing rows can change the population mixture

Group	Original rows	Missing rows	Rows remaining
A	80	4	76
B	20	10	10

Before deletion

GroupB is 20% of 100 rows. Missingness is 5% inA but 50% inB.

$$P(B \mid \text{complete}) = 10/86 \approx 11.63\%.$$

Consequence

The remaining table is cleaner but no longer represents the original group proportions. A model fitted only to complete rows may learn a different population.

Choice

Inspect missingness by relevant groups and validate the intended population. Low overall missingness alone does not justify complete-case deletion.

Mean and median encode different assumptions

A numeric column

Observed values 1, 2, 3, 100 give mean 26.5 and median 2.5. Replacing a missing value by the mean places it far from the typical small values.

Choice	Appropriate question
Mean	Is a symmetric numeric average meaningful?
Median	Is a skewed variable dominated by outliers?
Mode	Which observed category should a simple categorical baseline use?
Delete	Is missingness small enough and sufficiently unselective?

Information loss

Simple imputation reduces variation and ignores uncertainty. It is a baseline, not a recovery of the true missing value.

An extreme value changes mean imputation much more

Observed training values

$1, 3, 5, 100$ give mean $(1 + 3 + 5 + 100)/4 = 27.25$ and median $(3 + 5)/2 = 4$.

Two imputations

A missing entry becomes 27.25 under the mean rule or 4 under the median rule. Neither is an observed recovery of that entry.

Interpretation

The mean retains the large value's contribution to the average. The median describes the middle of the ordered sample and is less moved by this extreme observation.

Validity

First determine whether 100 is an error or a genuine rare case. Choosing the median does not justify deleting valid extremes.

The training median fills the new row

Training lab column

$(1, 3, \text{missing})$. Fit the median using observed training values, $m = (1 + 3)/2 = 2$.

$$\text{impute}(1, 3, \text{missing}) = (1, 3, 2), \qquad \text{impute}(\text{new missing}) = 2.$$

A missing indicator

Add $q = \mathbf{1}[\text{lab missing}]$. Training rows become lab/indicator pairs $(1, 0), (3, 0), (2, 1)$. The new missing row becomes $(2, 1)$.

Interpretation

The filled value allows numeric computation. The indicator retains whether an actual measurement existed.

Change over time

An ordering policy change can alter what missingness means. The indicator is useful only if the process remains relevant.

A missing indicator separates identical filled numbers

Rawlab	Filled lab	Missing indicator	Meaning
Measured 2	2	0	Observed value
Missing	2	1	Median replacement

Model information

Without the indicator, both records become the same lab coordinate. With it, a model can learn different associations for measured and unmeasured cases.

Process question

A clinician may omit a test for routine patients. The indicator can describe that decision process.

Transfer

If testing policy changes, the association can change even when biology does not. Evaluate whether the missingness signal persists in the intended setting.

Distribution and units guide the next operation

Before scaling

Check units, plausible ranges, skew, extreme values and subgroup mixtures. A large value can be a rare valid case or an entry error.

Unit inconsistency

Most heights are 1.5–1.9 metres. A value 170 may mean 170 centimetres. Dividing it by a standard deviation hides rather than fixes the unit mistake.

Histogram reading

A long right tail suggests checking log transforms. Two peaks can represent two populations. A spike at zero can represent an instrument detection limit.

Purpose

A Gaussian-looking histogram is not itself evidence of useful predictive information. Keep rare valid cases unless the task provides a reason to remove them.

Standardization makes the scale explicit

$$z_j = \frac{x_j - \mu_j}{s_j}, \qquad \mu_j = \frac{1}{n}\sum_i x_{ij}, \quad s_j = \sqrt{\frac{1}{n}\sum_i (x_{ij} - \mu_j)^2}.$$

Age calculation

Ages 20, 40, 60 give $\mu = 40$, $s = \sqrt{800/3} \approx 16.33$. Standardized ages are $(-1.225, 0, 1.225)$. New age 50 gives $z \approx 0.612$.

Meaning

A value 0.612 is 0.612 training standard deviations above the training mean. The transformation is affine and preserves ordering.

Details

This population-variance convention matches StandardScaler. A constant column has no variation and needs explicit handling. Standardization does not imply a normal distribution.

Scale changes distances and regularized scores

Distance example

Two patients differ by 10 years in age and 100 mg/L in a lab. Raw squared Euclidean distance is $10^2 + 100^2 = 10100$. The lab dominates because of its numeric unit.

After training scales

With age scale 10 years and lab scale 100 mg/L, the squared standardized distance is $1^2 + 1^2 = 2$.

Model effects

Distance-based methods and gradient updates depend on scale. A coefficient penalty also compares numeric coefficient sizes.

Tree connection

A strictly increasing rescaling preserves numeric ordering and possible training partitions for an ideal single-feature threshold tree. This does not make scaling irrelevant for every pipeline.

Minmax and robust scaling answer different needs

Method	Rule learned in training	Important property
Minmax	$(x - x_{\min})/(x_{\max} - x_{\min})$	Observed training range becomes 0–1.
Robust	$(x - \text{median})/\text{IQR}$	Uses central quantiles and resists some outlier influence.

Minmax example

Training range 20–60 maps 50 to 0.75. New 80 maps to 1.5 unless explicitly clipped. Refit-free extrapolation is expected.

Limits

Robust scaling does not delete outliers. Minmax does not guarantee every future value remains within 0–1. Specify zero-range or zero-IQR handling.

Robust scaling leaves a valid extreme visible

Five training values

$0, 1, 2, 3, 100$ have median 2. Under the stated quantile convention, $Q_1 = 1$, $Q_3 = 3$, so IQR2.

$$z_{\text{robust}} = (x - 2)/2: \quad 0 \mapsto -1, \quad 3 \mapsto 0.5, \quad 100 \mapsto 49.$$

Meaning

The centre and scale resist the extreme value, but the extreme remains extreme. Robust scaling does not remove or clip 100.

Details

Small-sample quantile conventions can differ. Save the exact training rule and handle zero IQR. Check whether extreme values are valid before deciding on a separate clipping policy.

Row normalization changes a different quantity

Column scaling

Fit a centre and scale for each feature across training examples. It retains each observation's relative magnitude.

Row normalization

$x \mapsto x/\|x\|_2$ fixes each observation's length. It emphasizes direction and removes overall magnitude.

Calculation

$(3, 4)$ has norm 5 and becomes $(0.6, 0.8)$. $(6, 8)$ becomes the same vector.

Suitable question

In a word-count vector, direction may describe topic while length reflects document size. For a patient's dose or total spending, magnitude may itself be predictive.

Validity

Handle the zero vector. Normalizing a row of incompatible units can create a quantity without a useful physical interpretation.

Column scaling and row normalization preserve different contrasts

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Two observations

$X = \begin{pmatrix} 1 & 10 \\ 3 & 30 \end{pmatrix}$. Columns have means 2, 20 and population scales 1, 10.

$$X_{\text{standardized}} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Row normalization

Both original rows instead map to $(1/\sqrt{101}, 10/\sqrt{101}) \approx (0.0995, 0.9950)$. Their common direction survives, but their magnitude difference disappears.

Choice

Use column rules when feature units need comparison. Use row normalization when composition rather than total amount is the intended signal.

A log transform expresses diminishing changes

$$z = \log(1 + x), \qquad x \geq 0 \text{ for the simple count example.}$$

x	0	9	99
$\log(1 + x)$	0	$\log 10 \approx 2.303$	$\log 100 \approx 4.605$

Meaning

The raw increases 9 and 90 become equal increments in transformed coordinates. Large values compress more strongly.

Advertising example

A model linear in log spending describes a smaller marginal effect at high spending than a model linear in raw spending.

Domain

Negative measurements need a justified transform. Scaling and distribution symmetry do not by themselves establish a better predictive relationship.

Log features can straighten a diminishing-return relationship

Spendx	Outcome	Raw endpoint line	Log-feature model
0	0	0	0
9	1	2/11≈0.182	1
99	2	2	2

Constructed example

$y = \log_{10}(1 + x)$. The raw line through endpoints is $\hat{y} = 2x/99$, while the log model is $\hat{y} = \log(1 + x)/\log 10$.

At 9

The raw straight line misses the intermediate outcome. A model linear in the log feature represents all three outcomes exactly.

Evidence

This deliberately defined example shows representation. Real advertising data need validation rather than assuming the log law.

Binning turns a numeric scale into interval features

Intervals

Age bins $[0, 30), [30, 50), [50, \infty)$ map 20, 40, 60 to three different categories. New 50 enters the third bin.

Linear model connection

One-hot bin indicators give each interval its own coefficient. A linear model can then express a piecewise constant age effect.

Quantile bins

Training quantiles choose approximately equal-count intervals. Store the learned boundaries and apply them unchanged to new rows.

Cost

Binning loses within-bin differences and creates abrupt boundaries. More bins increase dimensions and can leave sparse intervals. Compare it with a continuous baseline.

Quantile bins learn numerical boundaries from training data

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Ordered training incomes

$10, 20, 30, 40, 50, 60, 70, 80$. Using linear-interpolation quantiles gives $Q_{.25} = 27.5$, $Q_{.50} = 45$, $Q_{.75} = 62.5$.

Four training bins

The intervals contain $(10, 20)$,$(30, 40)$,$(50, 60)$ and $(70, 80)$, two observations each.

Future cases

Apply the saved boundaries. Income 46 enters the third bin even if a future batch's median is much higher. Define equal-boundary and out-of-range conventions explicitly.

Tradeoff

Equal training counts do not imply equal numeric widths or equal future counts. Binning sacrifices within-bin variation.

One-hot encoding avoids an invented category order

Ward	Indicator A	Indicator B	Indicator C
A	1	0	0
B	0	1	0
C	0	0	1

Nominal variable

Codes 1, 2, 3 would imply numeric distances and an order. One-hot columns let a model learn separate category effects.

Ordinal variable

A known severity order can use an ordinal code, but equal numeric spacing is still an extra modeling assumption.

Intercept

For an unregularized linear model with an intercept, all category indicators are linearly dependent. Dropping one reference category can resolve this. Different regularized models can use all indicators.

An intercept can duplicate the sum of one-hot columns

All three indicators

$I_A + I_B + I_C = 1$. With intercept 2 and coefficients $(1, 3, 0)$, predictions forA, B, C are 3, 5, 2.

Equivalent parameters

Intercept 1 and coefficients $(2, 4, 1)$ give the same predictions. Adding a constant to every category coefficient and subtracting it from the intercept changes no score.

Reference-category form

DropC and use intercept 2,$\beta_A = 1, \beta_B = 3$. The intercept representsC's score and remaining coefficients describe differences.

Scope

This identifies an unregularized parameter redundancy. Regularization and intercept handling can change which representation is preferable.

Unknown and frequent categories need explicit rules

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Training vocabulary

If fitted categories are A, B, then A maps to $(1, 0)$ and B to $(0, 1)$. With `handle_unknown="ignore"`, a new C maps to $(0, 0)$ in that category group.

Interpretation

All-zero unknown can share a representation with a dropped reference category. Use an explicit unknown group when the distinction matters.

High cardinality

A customer ID may create thousands of sparse columns. Rare-category grouping or feature hashing controls size. Hash collisions share a column.

Frequency encoding

If A occurs 8 times and B2 times, encode 0.8 and 0.2 using training counts. This describes frequency, not class probability.

A small hash space merges category identities

Category	Example bucket	Vector in 2 buckets
A	0	(1, 0)
B	1	(0, 1)
C	0	(1, 0)

Collision

A andC share the same representation, so the predictor cannot distinguish them through this encoded group alone. The mapping illustrates a hash result, not Python's particular randomized hash values.

Memory tradeoff

A fixed number of buckets keeps dimensions bounded as new identities arrive. More buckets reduce collision chance but increase storage.

Comparison

Rare-category grouping and target means lose different information. Choose with the actual model and validation task rather than calling every compact encoding equivalent.

Target encoding must exclude a row's own answer

Category target mean

$e_k = (n_k \bar{y}_k + \lambda \bar{y})/(n_k + \lambda)$ shrinks a rare category toward the overall training mean.

Leakage example

A unique ID appears once with label 1. Its unsmoothed target mean equals 1, revealing that row's answer to the predictor.

Training rows

Use out-of-fold encoding: each row's encoding comes from other folds. In time prediction, use only outcomes already known before the cutoff.

New rows

After choosing settings, fit category statistics on all allowed training labels and save them. An unseen category backs off to the saved global mean.

Limits

Adding noise alone does not remove the row's own label contribution.

Cross-fitting category means removes the row from its own statistic

Category	Fold 1 labels	Fold 2 labels	Full mean
A	1, 0	1, 1	0.75
B	0, 1	0, 0	0.25

Encode training rows

Fold 1 rows use Fold 2 statistics:A 1, B0. Fold 2 rows use Fold 1 statistics:A 0.5, B0.5. A row does not supply its own target to its encoding.

Final dictionary

For new rows, allowed full-training labels giveA0.75, B0.25. With smoothing $\lambda = 2$ and global mean 0.5, they become 2/3 and 1/3.

Evaluation

Outer validation labels stay out of both the cross-fitting and final dictionary.

A constructed feature answers a specific question

Raw measurements	New feature	Question answered
Total price, quantity	Price per unit	Is the item expensive for its size?
Height, weight	BMI	Is mass high relative to height?
Last visit, cutoff time	Days since last visit	How recent is the activity?
Two coordinates	Distance	How far apart are the locations?

Reasoning

A domain hypothesis motivates the construction. Units, zero denominators and availability define whether it is valid.

Evidence

A sensible story is a reason to test a feature, not proof that it improves prediction.

A physical ratio needs compatible units

BMI construction

For mass 70 kg and height 1.70 m, $\mathrm{BMI} = 70/(1.70)^2 \approx 24.22\ \mathrm{kg/m^2}$.

Unit error

Entering height 170 cm without conversion gives $70/170^2 \approx 0.002422$, a factor 10000 difference. Squaring the unit conversion matters.

Meaning

The ratio describes mass relative to squared height. It is a constructed measurement, not a complete clinical diagnosis. Raw height and mass may carry additional information.

Boundary

Confirm units, valid ranges and nonzero denominators before fitting. An apparently sophisticated formula cannot repair a wrong input definition.

Distance replaces four coordinates with a task quantity

Map example

For a local planar map with coordinates in km, pickup is $(1, 2)$ and destination $(4, 6)$.

$$d = \sqrt{(4-1)^2 + (6-2)^2} = 5 \text{ km}.$$

Why this helps

A simple fare model can use distance directly instead of discovering translation-invariant coordinate relationships from four columns.

Physical meaning

Straight-line distance is not road distance. Latitude and longitude require spherical or projected geometry, not an unqualified Euclidean formula in degrees.

Validation question

Compare a coordinate-only baseline with distance-added features on identical observations. Destination identity may still carry information beyond distance.

Ratios and differences represent different hypotheses

Unit price

100 yuan for 5 units gives 20 yuan/unit.200 yuan for 10 units has the same unit price but different total demand.

Change

A lab rises from 4 to 6 mg/L. Difference is 2 mg/L. Relative change is $(6-4)/4 = 0.5$.

Distinction

A ratio expresses size relative to a baseline. A difference retains an absolute unit. Keeping raw variables can retain magnitude lost by a ratio.

Small denominator

A change from 0.01 to 0.03 is a 200% increase but only 0.02 in absolute units. Use justified denominator handling and inspect noise sensitivity.

An interaction changes an additive model

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2.$$

Interpretation

The effect of x_1 depends on x_2: $\partial \hat{y} / \partial x_1 = \beta_1 + \beta_{12} x_2$.

Binary condition

Urgent message and unfamiliar sender are indicators. Their product equals 1 only when both are 1.

XOR construction

$z = x_1 + x_2 - 2x_1 x_2$ gives 0, 1, 1, 0 for input 00, 01, 10, 11. A linear threshold on z separates XOR.

Meaning

An interaction coefficient is a model term. It does not by itself establish that changing a cause would change an outcome.

A category cross names a specific combination

Country	Holiday	Country and holiday cross
CN	Spring Festival	CN $\times$ Spring Festival
CN	Ordinary day	CN $\times$ Ordinary day
US	Spring Festival	US $\times$ Spring Festival

Chinese-source motivation

A holiday may affect clicks differently in different countries. Separate additive indicators cannot automatically express every country-specific holiday effect.

Cross indicator

An indicator for CN $\times$ SpringFestival activates only for that pair, letting a linear model learn its additional effect.

Evidence

Rare combinations have few examples. A plausible interaction story still needs training support, regularization and held-out comparison.

Feature expansion grows faster than the evidence

Quadratic expansion

For p inputs, squares and pairwise interactions add $p(p+1)/2$ terms. Including linear terms gives $p + p(p+1)/2$ columns before an intercept.

Counts

$p = 2$ gives 5 columns: $x_1, x_2, x_1^2, x_1 x_2, x_2^2$. For $p = 100$, the same rule gives 5150 columns.

Sparse categories

Crossing 100 regions with 100 products creates 10000 possible combinations, many poorly observed.

Wide and deep connection

Explicit crosses store particular combinations. Learned dense embeddings can share information across related categories.

Practice

Choose useful constructions and compare regularized baselines. More columns do not imply more independent information.

A polynomial expansion has a calculable column order

Second-degree map without bias

$\phi(x_1, x_2) = (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$. For $(2, 3)$ this becomes $(2, 3, 4, 6, 9)$.

One new linear score

Set the interaction coefficient 1, other coefficients 0 and intercept -5. Then $a = 6 - 5 = 1$. In raw coordinates this is the nonlinear rule $x_1 x_2 > 5$.

Dimensions

Two original columns produce five feature columns. The model needs coefficients in that saved order.

Choice

Higher degree increases candidate terms quickly. A constructed map expands what the model can express, but it also increases fitting and validation demands.

Periodic time should preserve nearby hours

Calendar versus elapsed time

An absolute date can identify a trend. Days since the last event measure recency. Hour of day describes a cycle.

$$z_1 = \sin(2\pi h/24), \qquad z_2 = \cos(2\pi h/24).$$

Clock calculation

At 0 h, the encoding is $(0, 1)$. At 6 h it is $(1, 0)$. Hours 23 and 0 lie close in this two-coordinate representation.

Task

A cycle encoding removes any distinction between the same hour on different days. Keep appropriate date or elapsed-time information when it matters.

Availability

Compute all elapsed times relative to the actual prediction cutoff, not the date on which the dataset was exported.

Recency uses each observation's own prediction time

Prediction cutoff

At October 1, 12:00, the last known visit occurred September 29, 12:00. Recency is 48 hours.

Available versus future

A visit on September 30 can update that history if already recorded. A visit on October 2 is unavailable at the cutoff and must not enter the feature.

Two coordinates

Calendar date can represent season or trend. Elapsed hours represent how recently an event happened. They answer different hypotheses.

Consistency

Use the same timezone and each row's actual cutoff. Export-day subtraction would redefine old records after their decisions had already occurred.

Historical summaries need a closed information window

Prediction at time t

A seven-day purchase count uses events in $[t - 7\ \mathrm{days}, t)$ that were recorded by time t.

Row example

An event at $t - 1$ day is allowed. An event at $t + 1$ day is future information. An old event recorded only after t may also be unavailable.

Group aggregate

A user's past spending mean or a hospital's past rate can describe context. Use the history available before each row's cutoff.

Outcome delay

A past patient's outcome may become known later. Historical target statistics must respect the time when labels became available. Time-based validation should reproduce this boundary.

A rolling count must select events before it sums

Eventtime	Value	In 7-day window at t
$t - 8$ days	50	No
$t - 3$ days	20	Yes
$t - 1$ day	10	Yes
$t + 1$ day	40	No

Features at the cutoff

In $[t - 7\text{days}, t)$, count 2, total 30 and mean 15. The 40 from tomorrow cannot change any of these.

Label delay

A past event's eventual outcome is unavailable if it was not known by t. Occurrence time and recording time are both relevant.

Computation

Filter the valid history first, then aggregate. A full-table group mean can be a future statistic in disguise.

A useful construction must survive an ablation

Representation	Illustrative validation score	Meaning
Raw columns	0.80	Baseline
Raw + distance	0.84	Possible useful relation
Raw + distance + ID	0.85	Check split and availability

Ablation

Retrain with and without the constructed feature on identical folds. Compare the task metric, variation and acquisition cost.

No measured claim

These numbers illustrate comparison logic. An ID improvement may reflect memorized entities rather than transfer to new entities.

Transition

Construction expands candidate columns. Selection asks which columns deserve to remain, using a validation process that includes the selection itself.

Paired ablations separate mean gain from variability

Fold	Baseline	Addeddistance	Difference
1	0.81	0.83	0.02
2	0.82	0.83	0.01
3	0.80	0.79	-0.01
4	0.83	0.85	0.02
5	0.82	0.82	0.00

Same-fold comparison

Mean improvement is $(0.02 + 0.01 - 0.01 + 0.02 + 0)/5 = 0.008$. One fold worsens and one ties.

Conclusion

These illustrative results suggest a small average gain, not a guaranteed 0.8-percentage-point gain in every setting. Consider uncertainty and construction cost.

Search

If many constructions were tried, an outer evaluation must judge the chosen procedure.

Selection keeps columns, extraction creates coordinates

Operation	Output example	Consequence
Selection	Keep age and lab from a 10-column table	Original measurement meaning remains.
Extraction	$z_1 = 0.7x_1 + 0.7x_2$	New coordinate combines measurements.
Construction	Price/quantity	A task hypothesis defines a new variable.

Why select

Reduce acquisition cost, computation, noise and some redundancy. Preserve enough predictive information for the task.

PCA connection

PCA is an extraction method, treated in Lecture 11. Removing variables is not the same operation as projecting to principal components.

High dimension can dilute local evidence

Simple coverage calculation

Divide each unit-length feature into 10 intervals. Two dimensions create 100 cells. Ten dimensions create 10^{10} cells. A dataset of 1000 rows cannot populate most of them.

Consequences

Nearby samples can become rare, distance contrasts can weaken under some distributions, and chance associations become easier to find.

Boundary

High dimension is not automatically fatal. Relevant low-dimensional structure, sparsity, regularization and large sample sizes can help.

Selection objective

Discard variables that add little useful evidence while retaining interactions and complementary information. A low univariate score is not a universal deletion rule.

Three selection families use different evidence

Family	How it evaluates columns	Main cost or limitation
Filter	Variance or feature-target statistics	Fast, often misses interactions.
Wrapper	Retrained predictor on candidate subsets	Can reflect model performance, expensive search.
Embedded	Selection occurs during model fitting	Depends on the model and its regularization.

Examples

Correlation or mutual information is a filter. Forward selection and RFE are wrappers. L1 coefficients and tree splits support embedded selection.

Shared boundary

Every fitted selector belongs inside the training part of validation. A supervised filter is still using labels even before the final model fits.

Variance and correlation catch simple redundancy

Constant feature

A column with the same value in every training row has variance 0 and cannot separate those rows. A near-constant column may still mark an important rare event.

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}.$$

Unit conversion

Height in metres and height in centimetres have correlation 1. Keeping both adds no new measurement information.

Limits

Correlation measures linear association. $y = x^2$ on symmetric inputs can have correlation 0. Unsupervised redundancy removal still needs a justified choice of which measurement to retain.

A low-variance binary column can still identify a rare target

Rare indicator

A binary feature equals 1 for 1% of cases. Its population variance is $p(1-p) = 0.01(0.99) = 0.0099$.

Predictive counterexample

If the target is exactly this rare indicator, the feature perfectly predicts it despite the small variance. A threshold 0.01 would discard it.

Filter scope

Variance detects limited variation, not absence of target information. Constant columns and rare meaningful events require different interpretations.

Decision

Use task knowledge and held-out behavior before deleting rare-event variables. Compute fitted thresholds from allowed training data.

Zero correlation can coexist with a deterministic relationship

x	$y = x^2$	$x - \bar{x}$	$y - \bar{y}$
-1	1	-1	1/3
0	0	0	-2/3
1	1	1	1/3

Centres

$\bar{x} = 0, \bar{y} = 2/3$. The covariance numerator is $(-1)(1/3) + 0 + (1)(1/3) = 0$, so correlation 0.

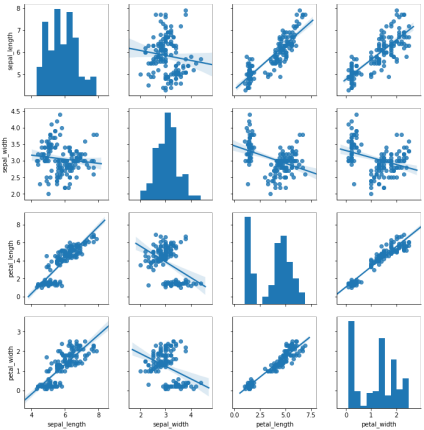
Meaning

The label is exactly determined by the input, but the increasing and decreasing sides cancel in the linear statistic. A square feature exposes the relationship.

Scope

Correlation filters measure linear association. They cannot certify general independence or rule out useful nonlinear predictors.

A pairplot shows distributions and dependencies



Panel order

Sepal length, sepal width, petal length, petal width. Rows and columns use this same order.

Reading

The diagonal shows individual distributions. Other panels show pairs of measurements.

Petal pair

The last two features rise together. This suggests shared information, but validation must test whether removing one loses useful predictions.

A second perfectly correlated column adds no new measurement

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Exact conversion

Let $x_2 = 100x_1$. Given x_1, the second column is known exactly. Both can individually have a strong relationship with the target.

Redundancy

Keeping height in metres and centimetres does not double the number of independent observations. Their coefficient values change with the unit convention.

Selection question

Choose an available, precise and maintainable representation. A second noisy measurement could add information, but an exact deterministic duplicate does not.

Model behavior

Duplicated inputs can alter penalties and numeric conditioning even when they add no measurement information. Validate the reduced pipeline.

Mutual information measures label dependence

$$I(X;Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)} = H(Y) - H(Y \mid X).$$

Binary example

Balanced X perfectly determines balanced Y: $I = 1$ bit. If X and Y are independent, $I = 0$.

What it detects

Dependence can be nonlinear. A feature such as X can inform a target $Y = X^2$ even when linear correlation is zero.

Estimation

Continuous-data scores need an estimator. Small samples and many candidate variables can create unstable or chance rankings. Compute scores using training labels only.

A binary mutual information score can be computed from counts

$X \setminus Y$	0	1
0	4	1
1	1	4

Probabilities

Ten cases give marginals 0.5 and joint probabilities 0.4 or 0.1.

$$I = 2\,[0.4 \log_2(1.6) + 0.1 \log_2(0.4)] \approx 0.2781 \text{ bits.}$$

Equivalent reading

$H(Y) = 1$ and $H(Y \mid X) \approx 0.7219$, giving the same decrease 0.2781. X improves knowledge ofY without determining it perfectly.

Estimate

This is an empirical score from ten observations. A ranking among many variables needs validation and stability checks.

Categorical tests compare observed and expected counts

Feature	$Y = 1$	$Y = 0$
Present	30	10
Absent	10	50

Independence expectation

Among 100 cases, 40 have the feature and 40 are positive. Expected present/positive count is $40 \cdot 40/100 = 16$, while 30 are observed.

$$\chi^2 = \sum_{\text{cells}} \frac{(O - E)^2}{E}.$$

Use of the score

Add discrepancies from all four cells. A large score challenges independence, but does not establish causation. Sklearn's chi-square selector requires nonnegative inputs, so standardized negative values violate that interface.

The chi-square example adds all four discrepancies

Cell	Observed	Expected	$(O-E)^2/E$
Present, +	30	16	12.2500
Present, -	10	24	8.1667
Absent, +	10	24	8.1667
Absent, -	50	36	5.4444

$$\chi^2 \approx 34.0278.$$

Expected counts

Each expected count equals its row total times column total divided by 100. The sign of $O - E$ disappears after squaring.

Interpretation

A large statistic challenges the stated independence model. It does not measure a causal effect or prove a feature will improve every predictor.

ANOVA compares between-group and within-group variation

Three classes

Feature values are $(1, 2, 3)$,$(5, 6, 7)$ and $(9, 10, 11)$. Group means are 2, 6, 10 and the overall mean 6.

$$SS_B = 3[(2-6)^2 + 0 + (10-6)^2] = 96, \quad SS_W = 2 + 2 + 2 = 6.$$

$$F = \frac{SS_B/(3-1)}{SS_W/(9-3)} = 48.$$

Meaning

Group centres are far apart relative to within-group variation. This numeric feature strongly separates the displayed group means.

Scope

ANOVA assumptions and sample design matter. The calculation is a filter statistic, not an independent estimate of final classification accuracy.

XOR shows why univariate ranking can fail

x_1	x_2	$y = x_1 \text{ XOR } x_2$
0	0	0
0	1	1
1	0	1
1	1	0

One column at a time

For either value of x_1, half the labels are 1. The same is true for x_2. Thus each feature alone has zero mutual information with y in this balanced example.

Together

The pair determines the label perfectly. Constructing an interaction or comparing joint subsets can preserve the signal.

Decision

A univariate filter can remove both useful columns. Zero individual score is not a proof of irrelevance in every model.

A useful word reduces label uncertainty

Word free	Spam	Not spam
Present	3	1
Absent	1	3

Eight documents

Labels are balanced, so $H(Y) = 1$. In each word-presence group, class proportions are 0.75 and 0.25.

$$I(\text{free}; Y) = 1 - H(0.75, 0.25) \approx 1 - 0.8113 = 0.1887 \text{ bits.}$$

Contrast

A word appearing equally across both classes provides no such evidence. Extremely rare words can receive unstable scores from very few documents.

Pipeline

Fit vocabulary and selection only on training documents. Evaluate the selected representation with the intended classifier.

Subset selection uses a predictor as its judge

Wrapper question

Which subset predicts well with this chosen learner? For each candidate subset, fit the predictor on training rows and score separate validation rows.

Search space

With p features, there are 2^p subsets. For $p = 20$, that is 1048576 candidates before any parameter tuning.

Model dependence

A linear model may need a constructed interaction. A tree may use the original pair conditionally. Their best subsets can differ.

Repeated selection

Trying many subsets can exploit validation noise. Use an outer evaluation loop or an untouched final test set to judge the completed search.

Forward selection follows measured increments

Candidate at current step	Illustrative validation score
Age alone / lab alone / ward alone	0.70 /0.76 /0.65
Lab+age / lab+ward	0.80 /0.77
Lab+age+ward	0.79

Search

Start with no features. Add lab, then age. Stop at lab+age if adding ward does not meet the improvement rule.

Limit

Greedy additions do not examine every combination. Pure interactions may fail to improve any one-feature candidate. Scores shown here illustrate the search, not actual experiments.

An added feature is judged relative to the current subset

Current	Candidate added	Example score
Lab	Age	0.80
Lab	Ward	0.77
Lab+Age	Ward	0.79

Conditional increment

Age wins after lab even if ward had a reasonable single-column score. The third row evaluates ward after both lab and age, not its original univariate association.

Search consequence

Two useful columns can be redundant. A weak individual column can become useful with a partner. Compare retrained subsets rather than adding standalone scores.

Limits

Forward search evaluates a restricted path through 2^p subsets. The displayed scores illustrate the logic and do not prove the globally best subset.

Backward elimination and RFE remove different candidates

Backward search

Start with all columns. Evaluate retrained subsets after removing one column and remove the least harmful candidate.

Recursive feature elimination

Fit a model that supplies coefficients or importance. Remove the lowest-ranked columns, refit and repeat until the desired count remains.

Coefficient example

Scaled columns age, lab, ward have coefficient magnitudes $(0.1, 0.8, 0.3)$. One-step RFE removes age, then refits on lab and ward.

Dependence

Rankings can change after removal. Coefficients of differently scaled columns are not directly comparable. The desired remaining count needs validation selection.

Backward search can remove a feature that once looked useful

Retrained subset	Example score
Age+Lab+Ward	0.80
Lab+Ward	0.78
Age+Ward	0.73
Age+Lab	0.83

One backward step

Removing ward produces the strongest candidate score 0.83, so the next subset is age+lab. The other removals degrade performance.

Interpretation

The removed variable can carry some association yet add variance or redundant details to this fitted model.

Evidence

These are illustrative validation results. Repeated search can overfit the validation set, so outer evaluation judges the whole removal procedure.

RFE refitting can change which feature ranks last

Fit	Active columns	Coefficient magnitudes	Remove
1	A, B, C, D	0.10, 0.20, 0.80, 0.70	A
2	B, C, D	0.60, 0.30, 0.05	D
3	B, C	Re-estimate	Stop at 2

Recursive meaning

After A is removed, the remaining model adapts. D becomes least important even though its original coefficient was large.

Contrast

Sorting the first coefficients once would next remove B. That is a different method from recursive elimination.

Settings

Use comparable feature scales and select the desired count inside validation. Magnitudes in this table are an illustrative sequence.

The search budget is also a modeling choice

Fit count

Selecting 5 features from 20 by basic forward search evaluates $20 + 19 + 18 + 17 + 16 = 90$ candidates. Five-fold scoring uses 450 predictor fits. Hyperparameter choices multiply this further.

Practical comparison

A quick filter can reduce an enormous vocabulary before a wrapper explores a smaller candidate set. An embedded sparse model can be a lower-cost baseline.

Fairness

Keep evaluation folds, preprocessing and search budgets comparable. More search can improve the best observed score while also increasing validation overfitting.

Decision

Use time and measurement cost alongside predictive uncertainty, especially when small score differences are unstable.

Nested validation evaluates the whole search

Outer training rows	Use an inner validation loop to choose preprocessing, feature count and model settings.
Outer validation rows	Apply the chosen fitted pipeline and score once for this outer fold.
Repeat outer folds	Summarize variability of the completed selection process.

Leakage contrast

Selecting the top 10 features using every label before creating outer folds lets each outer answer influence training.

Final model

After evaluation, run the specified selection procedure on all allowed training data. A final reserved test set remains separate.

Reading

Different outer folds can choose different columns. This is evidence about selection variability, not an implementation error.

Outer and inner folds have different responsibilities

Rows	Role	Labels may select features
Outer training 80	Split into 64 train and 16 validation	Yes, inside the inner procedure
Outer held-out 20	Evaluatechosenpipeline	No

One outer fold

Feature count and model settings are chosen using only the 80 outer-training rows. The 20 outer-held-out rows receive predictions from that completed choice.

Repeat

Rotate the outer split, then summarize the process's scores and chosen subsets. Different subsets across folds reflect limited-data variability.

Leakage

The 64/16 division is recreated within each outer training set. Never filter with all 100 labels first.

L1 can select while fitting the model

$$\min_{\beta_0,\beta} \frac{1}{2n}\|y-\beta_0\mathbf{1}-X\beta\|_2^2+\lambda\sum_{j=1}^{p}|\beta_j|.$$

Two objectives

The squared-error term fits the outcomes. The L1 term charges for coefficient magnitude and can set some coefficients exactly to zero. The intercept is usually unpenalized.

Selected columns

With coefficients $(0, 0.7, 0, -0.2)$, columns 2 and 4 remain active in this fitted linear model.

Settings

Scale features appropriately, choose λ in validation and compare with a dense baseline. Zero coefficients describe this model under this penalty, not universal irrelevance.

One-dimensional L1 fitting has an explicit threshold

Simplified standardized objective

$Q(\beta) = \frac{1}{2}(\beta - z)^2 + \lambda|\beta|$, where z is the unpenalized fitted coefficient in an orthogonal setting.

$$\hat{\beta} = \text{sign}(z)\max(|z| - \lambda, 0).$$

At $\lambda = 1$

$z = 3$ gives 2,$z = -2$ gives -1 and $z = 0.6$ gives 0.

Why zero appears

For positive β, the derivative gives $\beta = z - \lambda$; for negative β, it gives $\beta = z + \lambda$. If $|z| \leq \lambda$, the optimum is at the absolute-value kink 0.

Scope

Correlated multi-feature designs require joint optimization, not independent thresholding of arbitrary coefficients.

L2 shrinks coefficients without the same zero threshold

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Compare simplified objectives

L1 uses $\frac{1}{2}(\beta - z)^2 + \lambda|\beta|$. L2 uses $\frac{1}{2}(\beta - z)^2 + \frac{\lambda}{2}\beta^2$.

$$\hat{\beta}_{L2} = z/(1 + \lambda).$$

At $\lambda = 1$

$z = 3, -2, 0.6$ become $1.5, -1, 0.3$ underL2. UnderL1 they become $2, -1, 0$.

Selection distinction

The small nonzero 0.3 still uses its feature. A further cutoff would be an additional selection rule.

Convention

Penalty scaling varies by objective definition. Compare the stated formulas rather than assuming equal library alpha values mean equal regularization.

Coefficient size depends on units and substitutes

Same relationship

$\hat{y} = 2x$ with x in metres becomes $\hat{y} = 0.02x'$ when $x' = 100x$ in centimetres. Coefficient magnitude changes by 100 while predictions stay identical.

Penalty effect

L1 charges coefficient size. A meaningful scale must be defined before comparing penalized coefficients.

Correlated columns

Two near-duplicate measurements can substitute for each other. Lasso may keep one and discard the other, with the choice changing across samples.

Selection claim

A sparse model can be useful and unstable in its chosen names at the same time. Inspect validation performance and selection frequencies rather than treating one ranking as permanent.

Duplicate columns can exchange coefficients without changing predictions

Identical columns

If $x_A = x_B$, the score contribution is $(\beta_A + \beta_B)x_A$. Coefficients $(3, 0)$ and $(1.5, 1.5)$ produce identical predictions.

L1 cost

Both positive pairs have absolute-value sum 3. An L1 objective can be indifferent among such allocations in this exact duplicate setting.

L2 cost

Squared-coefficient sums are 9 and 4.5, so L2 prefers sharing the total in this simplified comparison.

Interpretation

Changing which name receives the coefficient need not change the learned relationship. Selection frequencies and acquisition choices should account for substitutes.

Lasso sets some linear coefficients to zero

Term	Least squares	Ridge	Lasso
TV	3.983944	3.981524	3.921642
Radio	2.860230	2.858304	2.806374
Newspaper	0.038194	0.038295	0.000000
Intercept	13.969091	13.969282	13.972528

What the source example shows

At the displayed penalty, Lasso uses TV and radio while setting the newspaper coefficient to zero. The intercept remains.

Changing the penalty

Different strengths produce different active columns. Validation chooses the strength. The numbers also depend on the dataset and preprocessing.

Interpretation

A zero coefficient in this fit does not establish universal irrelevance. Correlated substitutes can change which column remains.

Trees select conditions but importance needs validation

Embedded choice

A tree chooses features while choosing splits. A fitted Iris tree may use petal length and petal width while leaving two sepal measurements unused.

Training importance

Impurity importance can prefer features with many possible cut points. Correlated substitutes can hide one another.

Permutation test

On held-out rows, shuffle one column and compare the chosen prediction metric. This tests reliance of the fitted model, not causality.

Removal test

Retrain a pipeline without a candidate column and compare on identical folds. Acquisition decisions need the performance of the reduced system, not a training ranking alone.

Tree importance includes the fraction reaching each split

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Two useful splits

A root onfeatureA reaches 100 cases and decreases local Gini by 0.10. A child onfeatureB reaches 40 cases and decreases its local Gini by 0.20.

$$\text{contribution}_A = 1(0.10) = 0.10, \quad \text{contribution}_B = 0.4(0.20) = 0.08.$$

Normalized importance

With no other contributing splits, A receives $0.10/0.18 \approx 0.556$ andB $0.08/0.18 \approx 0.444$.

Training diagnostic

A larger local decrease does not necessarily mean larger whole-model contribution. Chance split opportunities and correlated substitutes still affect the ranking.

Permutation importance measures a fitted model's dependence

Shuffled column	Held-out score	Drop from 0.90
Age	0.88	0.02
Lab	0.78	0.12
ID	0.901	-0.001

One fitted predictor

Keep model parameters fixed and shuffle a column among held-out rows. The lab shuffle causes the largest displayed score drop.

Negative estimate

The tiny score rise after ID shuffle can reflect sample variation or an unhelpful dependency. Repeat permutations and examine uncertainty.

Different question

Retraining without a column lets the model adapt. It is a different experiment from shuffling, especially when other columns are correlated substitutes.

Selection stability includes cost and repeated evidence

Repeated fits

Across 10 training resamples, lab A appears 9 times, labB4 times and an ID2 times. A and B may substitute because they measure similar processes.

Interpretation

A frequency near 1 indicates stable selection under the specified procedure. It does not prove causality. A low frequency does not prove uselessness when substitutes exist.

Practical choice

Compare predictive variation, measurement cost, availability and subgroup performance. A two-test panel can be preferable to ten tests with an uncertain tiny score gain.

Many candidates

False discoveries grow when many unrelated columns are tested. Keep the full selection process inside validation.

Many noise features create opportunities for chance findings

Independent null tests

If 20 irrelevant features each have false-positive probability 0.05, the expected number flagged is $20(0.05) = 1$.

$$P(\text{at least one flag}) = 1 - 0.95^{20} \approx 0.6415.$$

Large vocabulary

For 1000 null features, the expected number at the same per-test rate is 50. A top-ranked feature can look impressive simply because many were examined.

Assumptions

The probability formula assumes independent tests. Real correlated features change it. Use appropriate multiplicity controls and evaluate the full selection process on unseen data.

Two selected sets can have stable scores and changing names

Two refits

Selected sets are $S_1 = \{A, B, C\}$ and $S_2 = \{A, B, D\}$. Their held-out scores are 0.89 and 0.90.

$$J(S_1, S_2) = \frac{|S_1 \cap S_2|}{|S_1 \cup S_2|} = \frac{2}{4} = 0.5.$$

Reading

Prediction scores are close, but only half the union of selected names is shared. C andD may be substitutes rather than unrelated random noise.

Decision

Inspect repeated selection frequencies, predictive variation, costs and availability. A stable reduced panel is valuable when its measurements can be used reliably.

A fitted pipeline preserves the complete calculation

```
num = make_pipeline(SimpleImputer(strategy="median"),
                    StandardScaler())
cat = make_pipeline(SimpleImputer(strategy="most_frequent"),
                    OneHotEncoder(handle_unknown="ignore"))
prep = ColumnTransformer([
    ("num", num, ["age", "lab"]),
    ("cat", cat, ["ward"])])
model = make_pipeline(prep, LogisticRegression(max_iter=1000))
scores = cross_val_score(model, X_train, y_train, cv=5)
```

Why inside the pipeline

Each fold learns preprocessing and the model from its training rows. A fitted selector belongs inside this pipeline.

Data for the code

Use a larger table with at least 5 observations per class. The three-row table is for hand transformations, not five-fold training.

Feature availability can veto a highly predictive column

Column	Known at admission	Eligible for this model
Age	Yes	Yes
Admissionlab	If measured before cutoff	Use with missing policy
Dischargedate	No	No
Finalbillingcode	No	No

Selection order

First define which information exists at the decision time. Then compare transforms and feature selection among legal candidates.

Prediction score

A future column can look excellent precisely because it reveals an outcome. High validation association cannot make an unavailable measurement usable.

Evaluation

A dataset created after events finish can contain future values for every training row. The task's timestamp boundary still applies.

One raw row becomes a vector with known coordinates

Saved rules from the three-row table

Lab median 2. Filled labs $(1, 3, 2)$ have mean 2 and scale $\sqrt{2/3} \approx 0.8165$. Age mean 40 and scale 16.33. Ward columns are A, B.

New row

Age 50, lab missing, ward C. With unknown-category ignore and a missing indicator, the feature vector is

$$\underbrace{(0.612,\ 0,\ 1,\ 0,\ 0)}_{\text{age z, lab z, lab missing, wardA, wardB}} .$$

Dimensions

Three raw fields produce five model columns. Save their order and meaning. The lab missing indicator is 1 even though the filled standardized lab is 0.

Boundary

The new row did not refit the median, mean, scale or vocabulary.

A saved feature vector produces a complete model score

Previously calculated vector

$z = (0.612, 0, 1, 0, 0)$ in order: age, lab, missing, ward A, ward B.

Illustrative logistic model

$\beta = (0.5, 1, -0.4, 0.3, -0.3)$ and intercept -0.2.

$$a = -0.2 + 0.5(0.612) + 1(0) - 0.4(1) + 0 + 0 = -0.294.$$
$$p = 1/(1 + e^{0.294}) \approx 0.427.$$

Prediction

A 0.5 threshold predicts the negative class. The missing indicator contributes -0.4 here. These illustrative weights show a calculation, not a clinical conclusion about unmeasured patients.

Selecting encoded columns changes the information retained

Coordinate	Value	Selected
Age z	0.612	Yes
Lab z	0	Yes
Lab missing	1	No
Ward A	0	No
Ward B	0	Yes

Mask and result

Mask $(1, 1, 0, 0, 1)$ gives $(0.612, 0, 0)$ in order: age, lab, ward B. Three raw fields created five columns, then selection retained three.

Lost information

Dropping the missing indicator makes an imputed mean lab and an observed mean lab identical through the retained numeric column. A selector acts on its encoded inputs. Validate whether this loss is acceptable.

A preprocessing script can leak before the classifier fits

Wrong order

A script concatenates training and test rows, fits a scaler, selects top features using all labels, then cross-validates only the classifier.

Two repairs

Fit each learned transformation on the fold's training rows. Fit supervised selection on those rows' labels. Score the resulting whole pipeline on the fold's validation rows.

After settings are chosen

Refit the complete specified pipeline on all allowed training data. Transform the reserved test with saved parameters and report its predictions.

Check

A clean classifier call cannot undo information already used by preprocessing or selection. The fitting boundary covers the entire system.

Exercise finds both numerical and information errors

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A: calculation

Training values 1, 3, 5 have mean 3 and population standard deviation $\sqrt{8/3}$. Standardize new value 7. Is its z-score above 1 an error?

B: leakage

A script target-encodes every row using every label, selects the best 10 columns, then runs cross-validation. Name both steps that leaked validation answers.

C: construction

Predict admission risk using “days until discharge”. Explain why this feature is invalid even if its formula is correct.

D: selection

Two XOR inputs each score 0 in a univariate filter. Should both always be deleted? Explain what a joint model can learn.

Answers connect representation to reliable learning

A

$z = (7 - 3)/\sqrt{8/3} = \sqrt{6} \approx 2.449$. It means 2.449 training standard deviations above the mean.

B

Fit target encoding without each training row's own label. Fit feature selection separately inside every validation fold. Evaluate the completed pipeline on separate outer or test data.

C

Discharge is future information at admission. Replace it with available history if that supports a meaningful hypothesis.

D

The pair determines XOR. Preserve justified interactions or evaluate suitable joint subsets rather than declaring both irrelevant.

Next lecture

We now know how inputs and search choices affect a model. Statistical learning theory asks how capacity, sample size and validation support generalization.